AMAZON BASE BUILDING COMPLIANCE STANDARD

Task Title	Electrical Installations
APM Object	CT.ELEC.ELECINST
APM Description	HV and LV wiring, switchgear, transformers and associated environments. Where sites have photovoltaic systems, items outside the scope of CT.ELEC.PHOTOV such as inverters and protection relays should be considered under this standard and/or CT.ELEC.FIXEDWIRE as appropriate.
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure. CT evidence must be additionally uploaded to the DMS (WorkDocs in DE) To suggest improvements please raise a SIM or contact document authors Leonardo Vitale (LEOVITAL) Rob Gisby (EWARTGXR) or APM, Benito Sanchez Sanchez de Puerta (BPPUERTA) SIM - Link to SIM template

Last Update: (Version)	abr-24	**Last Update Change Detail:**	PM evidence changed from 3M to 12M.	**Activities may be covered by No-Go restrictions:**	Not Specified	**Special Certification Required:**	**Delivered By:**
Evidenced part or full by PM:	Yes	**Related BBM PM:**	LEC.LV.BSBR ELEC..LV.CABS ELEC.HV.SWITCHGEAR ELEC.LV.PFC	**Related CT Areas:**	Not Specified		

Scope

Any site with an electrical installation. CEN and CENELEC standards define incoming electricity at:
Less than 1 kV - low voltage installations
More than 36 kV - high voltage
However, EU RME simplifies to LV < 1 kV AC and HV ≥ 1 kV AC.
Between 1 kV and 36 kV - medium voltage installation

Using this document

This document is intended as a guide and may not be exhaustive. Whilst this document is based extensively on professional guidance and subject to periodic review, individual sites must be diligent to ensure they are aware of, and comply to, local and changing regulations.
The format of these standards is intended to improve navigation between maintenance and compliance standards and to allow frequent, minor updates to a single living resource. Please avoid offline or printed versions and check version dates and change log.
In the event of becoming aware of errors or omissions in this document, sites must provide detail, including references to relevant source information to their local planner for escalation - SIM Link above.
This document is considered released as draft.
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Maintenance and Compliance Relationship

Where relevant, associated Base Building PM's are listed on the header . In many cases maintenance contributes to, or fulfils compliance requirement. This may not be exhaustive and is intended to assist users to identify relevant maintenance.
If a BBM Maintenance activity fully meets the compliance requirements below, periodic upload of PM evidence will be required to APM and DMS / Workdays depending on region.

Key to Abbreviations

The key to the below document is as follows:
CT - Amazon compliance / statutory task - not otherwise covered by a BBM Standard PM. Evidence to be provided into APM and DMS upon task closure.
CT* - This is a conditional task based or based on findings from previous inspections. If a frequency is not used, a further appropriate frequency needs to be in either the region, or 'ALL'. Local teams must ensure appropriate frequency.
(CT) - Compliance task which is likely met or exceeded by a CT task in [ALL] and therefore achieved by proxy BUT local site must satisfy themselves that requirements are fully met.
PM - Requirement can demonstrated through appropriate maintenance - Refer to relevant PM Standard
(PM) - Maintenance task which is met or exceeded by a PM task in [ALL] and therefore achieved by proxy BUT local site must satisfy themselves that requirements are fully met including regional / manufacturer requirements.
EVID - Periodic upload of evidence to DMS, for non-CT task lines. For example, evidence of PM completion, certification, licence etc.
EVID* - Conditional requirement to upload evidence at the interval provided
N A R - No Additional Requirement

Required Tasks

Region	Task Index	Task Frequency	Task Requirements	Qualification of Inspector	Records Required	PM	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	Additional
All	1	In accordance with manufacturer's and vendor's instructions and recommendations.	Routine maintenance of all electrical equipment and infrastructure	Competent, trained and certified vendor.	Maintenance logbook	PM					EVID		
All	2	Inspection frequency based on type of installation or equipment, its use and operating conditions (risk assessment), but at least every 3 to 5 years.	Thorough inspection of low voltage installations and equipment, including circuit breakers, interlocks, cover shutter devices, insulation, earthing, heating and lighting, testing, etc.	Competent, trained and certified vendor.	Inspection report, records or certificate. Risk assessment	PM					EVID		
All	3	Inspection frequency based on type of installation or equipment, its use and operating conditions (risk assessment), but at least every 3 to 5 years.	Thorough inspection of medium voltage installations and equipment, including circuit breakers, interlocks, cover shutter devices, insulation, earthing, heating and lighting, testing, etc.	Competent, trained and certified vendor.	Inspection report, records or certificate. Risk assessment	PM					EVID		
All	4	Inspection frequency based on type of installation or equipment, its use and operating conditions (risk assessment), but at least every 3 to 5 years.	Thorough inspection of high voltage installations and equipment, including circuit breakers, transformers, substations, busbars, insulation, earthing, switchgear, testing, etc.	Competent, trained and certified vendor.	Inspection report, records or certificate. Risk assessment	PM					EVID		
CZ	5	No additional requirements	Inspections as per [2], [3], and [4].	No additional requirements	Protokol o urceni vnejsich vlivu	(PM)					EVID		

DE	6	Every 3 years	Additional inspection to comply with building regulations (testing for effectiveness and operational safety).	expert according to state building law.	No additional requirements	(PM)					EVID		
DE	7	Based on insurance contract	Additional inspection (VdS-check) of HV installations and equipment for insurance cover purposes (fire risk)	VdS-approved electrical expert.	Periodic examination certificates VdS 2229	(PM)					EVID		
FR	8	Annual inspections.	Inspection as per [2], [3] and [4].	No additional requirements	No additional requirements	(PM)					EVID		
ROI	9	No additional requirements	Inspection as per [4]	No additional requirements	HV equipment register Schematic drawings of installation. Switchgear labelled.	(PM)					EVID		
UK	10	No additional requirements	Inspection as per [4]	No additional requirements	HV equipment register Schematic drawings of installation. Switchgear labelled.	(PM)					EVID		

Related Regional Legislation

Region	Relevant legislation	Vendor Certification	Statutory requirement
CZ	Decree 73/2010 Coll., ČSN 331500, ČSN 333320 ed.2, ČSN 332570, ČSN 332000, ČSN EN 62305, ČSN 331600.ed.2, Decree 50/1978	Competent electrician. Review technician.	Frequency determined by the Protocol on the list of external influences, between 6 and 60months.
DE	DGUV Vorschrift 3 (§5) - Unfallverhütungsvorschrift Elektrische Anlagen und Betriebsmittel DIN VDE 0105-100 - Betrieb von elektrischen Anlagen DIN VDE 0100-600 VDE 0100-600:2017-06 - Errichten von Niederspannungsanlagen Teil 6: Prüfungen VdS 2871 - Richtlinien für die Prüfung elektrischer Anlagen - Prüfrichtlinien nach Klausel SK 3602 - Hinweise für den anerkannten Elektrosachverständigen insurance contract	Qualified electrician (Elektrofachkraft).	Inspection must be conducted before first use and after any modification or repair. Inspection frequency based on risk assessment, but at least every 4 years for electrical installations and fixed electrical equipment, and every 6 - 24 months for electrical installations and mobile electrical equipment in special types of premises according to DGUV 4. Monthly functional inspection of residual current, mobile devices. Functional test of the residual and differential current, and residual voltage circuit breakers by user: daily for mobile equipment and every 6 months for fixed equipment.
ES	RD 842/2002, of August 2, which approves the electrotechnical regulation for low voltage and its 52 Complementary Technical Instructions. RD 337/2014, of May 9, which approves the high voltage electric systems and its 23 Complementary Technical Instructions.	Accredited company (OCA) by ENAC.	Low voltage Inspections and testing at least every 5 years. HV inspections and testing to be conducted at least every 3 years.
FR	Section 5 : Vérification des installations électriques (Articles R4226-14 à R4226-21) - Légifrance (legifrance.gouv.fr) Arrêté du 26 décembre 2011 relatif aux vérifications ou processus de vérification des installations électriques ainsi qu'au contenu des rapports correspondants - Légifrance (legifrance.gouv.fr)	Accredited organisation or employee who is qualified or trained.	Annual inspection
IT	Legislative Decree 8/04/2008 n.81 DPR 462/01 DM 22/01/2008 n.37 CEI EN 50110-1 (Operation of electrical installations) CEI 11-27 (Work on electrical installations) CEI 0-14 (Guidelines for application of DPR 462/01) CEI 64-14 (Guidelines for electrical installations inspections) CEI EN 60079-17 (Electrical installations maintenance for ATEX) CEI O-10 (Guidelines for maintenance) CEI 0-15 (Maintenance for medium voltage installations)	Competent person. Except for grounding systems inspection: National authorities (ASL, ARPA) or accredited inspection organization.	Routine maintenance. Grounding system inspection every 5 years after first put into service, or every 2 if there is a risk of fire. Electrical systems inspection every 5 years, or every 2 if there is a risk of fire.
PL	Act of 7 July 1994 of Construction Law (Journal of Laws 2020, item 1333) Regulation of Minister of Infrastructure of 12 April 2002 on technical conditions to be met by buildings and their location. (Journal of Laws 2019, item 1065) Act of 29 August 2014 on the energy performance of buildings (Journal of Laws 2021, item 497) Act of April 10, 1997 - Energy Law (Journal of Laws of 1997, No. 54, item 348). Regulation of the Minister of Economy, Labor and Social Policy of 28 April 2003 on the detailed rules for confirming qualifications by persons involved in the operation of devices, installations and networks (Journal of Laws of 2003, No. 89, item 828). Regulation of the Minister of Energy of August 28, 2019 on health and safety at work with energy devices (Journal of Laws 2019, item 1830).	Qualified or competent person	Inspection frequency depends on manufacturer's or vendor recommendations, but at least every 5 years.

ROI	S.I. No. 299/2007 - Safety, Health and Welfare at Work (General Application) Regulations 2007 (irishstatutebook.ie) NSAI 5th Edition National Rules for Electrical Installations, IS 10101:2020 on 6 March 2020.	Competent and qualified electrician, approved by RECI or ECSSA.	Routine maintenance. Frequency of inspection and testing depends on the type of installation, its use and operation, but at least every 3 to 5 years.
SK	Act No. 124/2006 Coll. on safety and health at work Act No. 314/2001 on fire prevention Ordinance No. 726/2002 Coll. on the characteristics of the electric fire-safety system, the conditions of its operation and ensuring its regular inspection Ordinance No. 508/2009 Coll. on details for ensuring safety and health protection at work with pressure, lifting, electrical and gas technical equipment	Qualified & certified person	Once a year in case of complex inspection. Otherwise quarterly, monthly, or daily.
UK	The Electricity at Work Regulations 1989 (legislation.gov.uk) Approved code of practice BS 6626: 1985 - Maintenance of Electrical Switchgear and Control gear for Voltages above 1 kV and up to and including 36 kV. BS 7671 / IET Wiring Regulations Keeping electrical switchgear safe HSG230 (hse.gov.uk) Electrical switchgear safety: A guide for owners and users (hse.gov.uk) The Electricity At Work Regulations (Northern Ireland) 1991 (legislation.gov.uk)	Registered electrician, approved by NICEIC, ECA or SELECT.	Routine maintenance. Inspection frequency of LV, MV and HV installation and/or equipment based on type of installation or equipment, its use and operating conditions, but at least every 3 (industrial premises) to 5 years (commercial premises)